

Smart Attendance with QR Verification and Live Session Integration for Hybrid Classrooms

Research Question to be answered

How effective, secure, usable, and privacy-preserving is a QR-based attendance system with Zoom auto-join compared to traditional methods in hybrid/in-person classes?

Overview, Justification and overall aim of project

- Manual roll calls are slow, error-prone, and unscalable.
- QR workflows are low-cost and device-friendly but vulnerable to code sharing.
- Live session links (Zoom) tie attendance to synchronous participation in remote or hybrid teaching.
- Institutions require ethical, privacy-preserving attendance systems without biometrics.

Justification

QR plus authenticated joins can balance integrity and usability. Few studies rigorously evaluate non-biometric attendance integrity under realistic constraints. This work addresses that gap.

Overall Aim

Design, implement, and evaluate a QR-based attendance system with Zoom auto-join, focusing on integrity, latency, usability, accessibility, and privacy, and produce deployment best practices.

Objectives

- Design architecture integrating sessions, rotating/per-user QR tokens, and Zoom links.
- Implement teacher workflow to start sessions with Zoom URL/ID and student auto-join.
- Instrument logging for integrity, latency, and usability metrics.
- Evaluate anti-sharing strategies (token lifetimes, per-user tokens, authenticated joins).
- Assess usability, accessibility, and privacy; develop deployment guidelines.
- Produce technical and policy recommendations for institutions.

Methodology

Approach

Experimental design comparing QR variants: static codes, rotating codes, and per-user tokens; mixed methods combining quantitative metrics and qualitative usability feedback.

Data and Collection

- Quantitative: timestamps, device type, network quality, join outcomes, duplicate/fraud flags.

- Qualitative: SUS questionnaire and short interviews on ease-of-use and barriers.

Analytical Tools

- Statistical tests (chi-square/ANOVA) to compare variants.
- Descriptive analytics dashboards for latency and integrity trends.
- Thematic analysis of qualitative feedback.

Ethical and Practical Issues

- Privacy-by-design: minimal PII, no biometrics, consent, retention limits.
- Accessibility: alternate flows for students without smartphones or stable internet; manual fallback with audit.
- Practical constraints: Wi-Fi variability and device heterogeneity; retries, grace windows, offline QR display.

Work Plan (13 Weeks)

- Week 1: Requirements, ethics prep, consent; finalize study design.
- Week 2: Implement sessions with Zoom URL/ID and student auto-open; baseline QR flow.
- Week 3: Add rotating QR tokens; lifetimes and join windows; logging.
- Week 4: Implement per-user tokens; authenticated join; anomaly rules.
- Week 5: Pilot in one class; collect integrity/latency logs; refine UX.
- Week 6: Broader pilot; SUS survey; accessibility checks.
- Week 7: Data cleaning; preliminary statistical analysis; adjust policies.
- Week 8: Usability interviews; thematic coding; incorporate quick wins.
- Week 9: Remote/hybrid evaluation; measure Zoom auto-join impacts.
- Week 10: Comparative analysis of variants; dashboards and tables.
- Week 11: Draft technical guidelines and privacy policies.
- Week 12: Finalize results, limitations, recommendations; internal review.
- Week 13: Prepare and submit final report; dissemination demo.

Relationship to BS Programme/Stream

- Software Engineering: backend/frontend design and integration.
- Web Systems: secure session and token management.
- HCI: usability and accessibility evaluation.
- Information Security: anti-sharing strategies and auditability.
- Data Science: experimental design and analysis.

Indicative Reading List and Resources

Indicative Reading List

- ISO/IEC 29100. Privacy framework for ICT systems.
- Nielsen, J. Usability Engineering.
- Brooke, J. SUS: A quick and dirty usability scale.
- OWASP Cheat Sheet Series: Authentication and Session Management.
- Boniface et al. Attendance systems in higher education: technology-mediated approaches.
- Zimmerman. Hybrid learning design and student engagement.

Resources

- Hardware: instructor laptop; classroom display; student devices.
- Software: FastAPI, Next.js, SQLite/PostgreSQL, Zoom, analytics, survey tools.
- Infrastructure: secure hosting, HTTPS, optional email notifications.